

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (CE) Examination

April 2013

CE 207 Java Programming

Date: 23.04.2013, Tuesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Do as Directed.

[11]

State True or False with justification.

- 1) A .class file contains byte codes.
- 2) The modifiers public and static cannot write in either order "public static" or "static public".
- 3) Interface can be instantiated.
- 4) Constructor overloading is not possible in Java.
- 5) The JRE deletes objects when it determines that they are no longer being used. This process is called Garbage Collection.
- 6) In an instance method or a constructor, "this" is a reference to the current object.
- 7) Java supports both multi dimension and nested arrays.

What is the output of the followings?

- 8) Consider the following code snippet

```
String river = new String("Columbia");  
System.out.println(river.length());  
What is printed?
```

- | | |
|------|-------------|
| a) 6 | b) 7 |
| c) 8 | d) Columbia |

- 9) What is different between a Java applet and a Java application?

- | | |
|---|---|
| a) An application can in general be trusted whereas an applet can't | b) An applet must be executed in a browser environment. |
| c) An applet is not able to access the files of the computer it runs on | d) (A), (B) and (C). |

Q - 2 Answer the Following Questions.

- (a) What is the meaning of Object Oriented Programming? List out the different languages which follow Object Oriented Concept. How it is differ from Procedure Oriented Programming?
- (b) Draw the architecture of JAVA. Expalin each part of its.

[06]

[06]

OR

Q - 2 Answer the Following Questions.

- (a) What is JVM? How it is important for JAVA? What is Byte Code? What is JDK? [06]
List out the four class libraries those you used in JAVA.
- (c) Explain Type Conversion with respect to expressions & assignments with suitable example. [06]

Q - 3 Answer the Following Questions.

- (a) Explain the following terms: [06]
1) Method Overloading
2) Polymorphism
3) Wrapper Classes
- (b) What is Inheritance? List out how many types of inheritance will be supported by JAVA. What is the use of constructor in JAVA? Which are the three uses of super() keyword? [06]

OR

Q - 3 Answer the Following Questions.

- (a) What is Interface and Package? Explain Interface Reference with example. [06]
- (b) Which are ways to set the classpath? Write steps to create and import package with example. [06]

SECTION - II

Q - 4 Do as Directed.

(a) **Fill in the Blanks.**

- 1) True and false values are represented by the _____ data type.
2) Java statements end in _____.

State True or False with justification.

- 3) public static final main (String[] args) {} is a valid method.
4) Can constructor throw exception?
5) GC is a high priority thread.

What is the output of the followings?

- 6) What is byte code in the context of Java?
a) The type of code generated by a Java compiler b) The type of code generated by a Java Virtual Machine
c) It is another name for a Java source file d) It is the code written within the instance methods of a class.
e) It is another name for comments written within a program.
- 7) sleep(50);
The above statement of a thread implies that the thread will get the processor-time
a) any moment after 50 milliseconds b) exactly after 50 milliseconds
c) even before 50 milliseconds if the processor is idle d) None
- 8) Which of the following statements about arrays is syntactically wrong in JAVA?

- a) Person[] p = new Person[5]; b) Person p[5];
c) Person[] p []; d) Person p[][] = new Person[2][];

Q - 5 Answer the Following Questions.

- (a) Explain the Life Cycle of Applet. What is the use of paint () method? Which method is used to print the text in applet? [03]
- (b) How thread is useful in JAVA? List the possible ways to create threads. Explain the Life Cycle of Thread. [05]
- (c) Explain exception handling. Can we have the try block without catch block? What is the difference between throw and throws? [04]

OR

Q - 5 Answer the Following Questions.

- (a) How I/O is performed in Java? Write a simple code which copies content from one binary file to another binary file. [03]
- (b) Explain the File and FileInputStream with one example. [05]
- (c) Why Layout Managers is needed to create the AWT programs? List out the different Layout Managers. Explain the Grid Layout in detail. [04]

Q - 6 Write down the Following Programs with output.

- (a) Write a program that creates and initializes a four element double array. Calculate and display the average of its values. [04]
- (b) Write an application that accepts two doubles as its command-line arguments, multiplies these together, and display the product. [04]
- (c) Write an applet that draws different shapes. Also fill those shapes. [04]

OR

Q - 6 Write down the Following Programs with output.

- (a) Write a program that calculates the area of a circle, rectangle, square and triangle as its command-line argument and display the area of these shapes. [04]
- (b) Write a program for multiplication of two 3*3 matrix where values has been taken from the input. [04]
- (c) Write a program that generates a custom exception if any of its command-line arguments are negative. [04]
